



## OSA Data Assessment Report

Case SiDa 1960

|                  |           |  |         |  |                  |                                    |
|------------------|-----------|--|---------|--|------------------|------------------------------------|
| Personal details | Gender: F |  | Age: 56 |  | Weight & Height: | 68 kg (150 lbs)<br>162 cm (5'4.2") |
|                  |           |  |         |  | BMI:             | 25.9                               |
|                  |           |  |         |  |                  |                                    |

**Clinical background:** Hypertension, allergies: Nickel sulfate, Palladium chloride, Sodium tetrachloropalladate, CPAP intolerance

**Patient presentation:** Reports include loud snoring at night ("I sleep heavily and don't feel it. The snoring is strong and disturbing to family members"), unrefreshing sleep ("wake up and immediately feel tired"), and excessive daytime sleepiness ("easily fall asleep during the day"). Difficulty falling back asleep after nighttime awakenings ("if I wake up in the middle of the night, it's hard to fall back asleep") and unintentional daytime napping. Occasional difficulty staying awake while driving, especially during long drives or at night/early afternoon.

TMJ-related symptoms include nighttime teeth grinding, for which she was prescribed a night guard that has helped significantly ("received a night guard that helps with bruxism"), and past headaches in the morning that have improved with consistent use of the night guard ("used to have morning headaches, now less frequent").

The patient previously attempted CPAP therapy but was unable to tolerate it due to shortness of breath and anxiety.

**Patient goals:** Snoring cessation, improved daytime fatigue, enhanced sleep quality, increased daily energy, reduced health risks, reduced TMJ-related pain, fewer morning headaches, and reduced teeth clenching or grinding.

### Sleep Study Data

|                  |                      |
|------------------|----------------------|
| Sleep Duration   | 6h and 09min         |
| AHI              | 10.7 (Mild OSA)      |
| O2 Nadir         | 87%                  |
| ODI              | 3.3                  |
| Less than 90% O2 | 0.1%                 |
| Snoring          | mean 41 dB, max 53dB |
| Sleep efficiency | 93.5%                |

### Observations:

- Data indicate mild OSA (AHI 10.7), with oxygen desaturation to a nadir of 87% and minimal time spent below 90% saturation.
- Desaturation events were recorded with an ODI of 3.3, consistent with intermittent hypoxic stress.
- Snoring was present, with a mean intensity of 41 dB and peaks up to 53 dB.
- Total sleep time was 6h09min, with a high sleep efficiency of 93.5%.

### Structural Observations from Imaging Data



| Key Observations                | Details                                                                       |
|---------------------------------|-------------------------------------------------------------------------------|
| <b>Obstruction Sites</b>        | Tongue base-related oropharyngeal narrowing                                   |
| <b>Soft Palate &amp; Uvula</b>  | Elongated and swollen, partially obstructing the oropharynx.                  |
| <b>Tongue Position</b>          | Large, posteriorly positioned tongue base encroaching the airway.             |
| <b>Bite &amp; Jaw Structure</b> | Cannot assess bite; narrow jaw arches observed.                               |
| <b>Hyoid Bone</b>               | Not fully visualized, but appears positioned inferior to the mandibular plane |
| <b>Nose &amp; Sinuses</b>       | Soft tissue density observed in the right maxillary sinus                     |

**Conclusion:** The provided CBCT imaging suggests potential airway obstruction primarily due to anatomical factors — including soft tissue hypertrophy of the soft palate and uvula, posterior tongue positioning, and oropharyngeal narrowing related to tongue base anatomy. Narrow jaw arches may further contribute to reduced airway volume. The hyoid bone appears to be positioned below the mandibular plane, which may reflect compromised airway support. Additionally, a soft tissue density observed in the right maxillary sinus.

### Possible Treatment Considerations

- History of CPAP intolerance, with reported shortness of breath and anxiety, supports consideration of alternative therapies.
- Mandibular advancement using an oral appliance may help stabilize the upper airway by repositioning the tongue base forward and away from the oropharyngeal wall — potentially reducing obstruction and improving airflow during sleep.
- Given the patient's known allergies to certain metals (e.g., nickel and palladium compounds), material selection for the oral appliance should avoid these components.

- Anterior and posterior vertical opening may contribute to improved airway patency by increasing space for posterior tongue repositioning and distancing the soft palate from the posterior pharyngeal wall.
- Intraoral scans revealed an anterior open bite, which may affect appliance retention and should be taken into account during device selection and design.
- Given the patient's history of bruxism and prior response to a night guard, any oral appliance should incorporate TMJ-friendly features and be carefully titrated to minimize stress on the masticatory system.

#### Device Design Data Considerations

| Parameter              | Data-Based Consideration                                             |
|------------------------|----------------------------------------------------------------------|
| Starting Point         | +2mm anterior from current IC position                               |
| Vertical Opening       | 5mm anterior                                                         |
| Anterior Window        | Large, allowing tongue access to upper incisors                      |
| Stabilization Features | Elastic hooks for stabilization placed at the cuspids area           |
| Material               | Nylon                                                                |
| Coverage               | Full coverage                                                        |
| Midline Correction     | Not needed (no deviation during protrusion)                          |
| Clinical Notes         | Avoid metal components containing nickel or palladium due to allergy |

#### Recommendations for Further Evaluation

- Initiate treatment with a mandibular advancement device tailored to the patient's airway anatomy and history of bruxism. Follow-up symptom tracking using the VizBriz follow-up questionnaire is recommended 4–6 weeks after initial fitting.
- Consider repeat sleep testing with the device in place after the titration phase (typically 8–12 weeks), to evaluate treatment effectiveness and guide further adjustments if needed.
- Refer to ENT specialist for evaluation of the soft tissue density observed in the right maxillary sinus.
- Monitor TMJ-related symptoms and joint function throughout treatment, including assessment of jaw pain, joint noises, and bite changes. Follow-up evaluation may be required if symptoms worsen or new issues arise.

## Oral Appliance Options for Consideration

**OrthoApnea NOA**



**Panthera X3**



**EMA 3D**



### **Disclaimer**

This AI-generated report is intended as a data assessment tool to support clinical decision-making. It does not constitute a medical diagnosis, treatment recommendation, or radiological interpretation. All imaging findings, including CBCT references, should be reviewed and confirmed by a certified radiologist or qualified healthcare provider. Final treatment decisions must be made by licensed professionals based on a comprehensive clinical evaluation. No liability is assumed for medical decisions made solely on the basis of this report.